

GBU 2301: Business Statistics

Week 4: Introduction to Probability

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Required reading: Anderson et al., Statistics for Business and Economics, 15th ed., Chapter 4

Outline

- Random Experiments and Counting Rules
- Assigning Probabilities and Events
- Basic Relationships of Probability
- Conditional Probability and Independence
- Bayes' Theorem

Random Experiments and Sample Spaces

- A random experiment produces one uncertain outcome. The sample space $\mathcal{S}$ contains all possible outcomes; an event is a subset of $\mathcal{S}$.
- For an order, record whether delivery is late (L) and whether a complaint occurs (C).

$$\mathcal{S} = \{(L, C), (L, C^c), (L^c, C), (L^c, C^c)\}$$

- A^c is the event that A does not occur.
- The four pairs are possible outcomes.
- The pairs need not have equal probabilities.

Random Experiments: Examples

Experiment	Possible outcomes
Toss a coin and record the upward face	Head, tail
Inspect a part for defects	Defective, nondefective
Make a sales call and record whether a purchase occurs	Purchase, no purchase
Roll a six-sided die and record the upward face	1, 2, 3, 4, 5, 6
Record one football team's result after regular time	Win, loss, draw

For the part inspection, $S = \{\text{defective, nondefective}\}$.

Two possible outcomes do not imply that each has probability $1/2$.

Counting Rule for Multiple Steps

For k steps with $n_1, n_2, \dots, n_k$ choices, the number of outcomes is $n_1 n_2 \cdots n_k$.

- A sales offer has three channels, two delivery options, and two payment methods.

$$3 \times 2 \times 2 = 12 \text{ possible combinations}$$

- Choose one option at each step.
- Multiply the number of choices available at each step.

Counting Rule: Coin Tosses

Toss a coin twice and record the result of each toss in order. Let H denote a head and T a tail.

$$S = \{(H, H), (H, T), (T, H), (T, T)\}$$

$$\text{Number of outcomes} = 2 \times 2 = 4$$

(H, T) and (T, H) are different outcomes because the first and second tosses are recorded separately.

For six tosses, recording the full sequence gives

$$2^6 = 64 \text{ possible sequences.}$$

If only the number of heads is recorded, the possible values are 0, 1, 2, 3, 4, 5, 6: seven values, rather than 64 sequences.

Counting Rule: Project Duration

Beta Energy plans a power-plant expansion with two successive stages. Design can take 2, 3, or 4 months; construction can take 6, 7, or 8 months. Assume every listed design time can be followed by every listed construction time.

How many outcome pairs are possible, and what are the possible total completion times?

Each cell below gives the total time in months.

Design time	Construction: 6 months	Construction: 7 months	Construction: 8 months
2 months	8	9	10
3 months	9	10	11
4 months	10	11	12

$$\text{Number of outcome pairs} = 3 \times 3 = 9$$

There are five distinct total times: 8, 9, 10, 11, and 12 months. For example, (2, 8), (3, 7), and (4, 6) all give 10 months.

Six of the nine pairs meet the target of completion within 10 months. Their probabilities are not specified.

Counting Rule: Investment Outcomes

An investor holds shares in Gamma Energy and Delta Mining. Their possible gains or losses after three months are:

Investment	Possible gains or losses, thousands of dollars
Gamma Energy	10, 5, 0, -20
Delta Mining	8, -2

Assume all combinations of the listed outcomes are possible. How many outcome pairs and distinct combined gains or losses are possible?

$$\text{Number of outcome pairs} = 4 \times 2 = 8$$

Each row below is one possible pair of outcomes. All amounts are in thousands of dollars.

Counting Rule: Investment Outcomes

Gamma Energy	Delta Mining	Combined gain or loss
10	8	$10 + 8 = 18$
10	-2	$10 + (-2) = 8$
5	8	$5 + 8 = 13$
5	-2	$5 + (-2) = 3$
0	8	$0 + 8 = 8$
0	-2	$0 + (-2) = -2$
-20	8	$-20 + 8 = -12$
-20	-2	$-20 + (-2) = -22$

The eight pairs produce seven distinct totals:

$$\{-22, -12, -2, 3, 8, 13, 18\}.$$

The pairs $(10, -2)$ and $(0, 8)$ both produce a total gain of 8.

Permutations and Combinations

Select r objects from n distinct objects, where $0 \leq r \leq n$.

Counting rule	Does order matter?	Number of selections
Permutation	Yes	$P(n, r) = \frac{n!}{(n-r)!}$
Combination	No	$C(n, r) = \binom{n}{r} = \frac{n!}{r!(n-r)!}$

Each unordered selection has $r!$ possible orders: $P(n, r) = r!C(n, r)$.

Permutations and Combinations

- A firm has six candidates.
- Appoint a lead and a backup. The roles make order matter:

$$P(6, 2) = \frac{6!}{(6-2)!} = 6(5) = 30$$

- Select a three-person committee. Rearranging the same three people does not form a new committee:

$$C(6, 3) = \binom{6}{3} = \frac{6!}{3!(6-3)!} = 20$$

Permutations and Combinations: Example

A firm has three projects: Alpha, Beta, and Gamma. It selects two different projects.

- Combination: Which two projects are selected?
- Permutation: Which project is presented first, and which is presented second?

Selected pair: order does not matter	Possible presentation orders: first, then second
Alpha and Beta	(Alpha, Beta), (Beta, Alpha)
Alpha and Gamma	(Alpha, Gamma), (Gamma, Alpha)
Beta and Gamma	(Beta, Gamma), (Gamma, Beta)

There are three pairs, but six presentation orders:

$$C(3, 2) = \frac{3!}{2!1!} = 3, \quad P(3, 2) = \frac{3!}{1!} = 6$$

Each selected pair has two possible orders:

$$P(3, 2) = 2!C(3, 2) = 2(3) = 6$$

Permutations and Combinations: Inspection

An inspector has five distinct parts, A, B, C, D, E , and selects two without replacement.

- If only the selected pair matters, $\{A, B\}$ and $\{B, A\}$ are the same selection.

$$C(5, 2) = \frac{5!}{2!3!} = \frac{5(4)}{2(1)} = 10$$

- If the first and second inspection positions are recorded, (A, B) and (B, A) are different arrangements.

$$P(5, 2) = \frac{5!}{3!} = 5(4) = 20$$

Each pair has $2! = 2$ possible orders, so $20 = 2(10)$.

Permutations: Examples

Product Codes

Beta Retail forms a three-letter product code using *A, B, C, D, E, F*. No letter can be repeated, and different letter orders give different codes.

$$P(6, 3) = \frac{6!}{3!} = 6(5)(4) = 120$$

Project Roles

Gamma Consulting assigns a project lead, a reviewer, and a presenter from eight employees. Every employee is eligible for every role, and each person can hold only one role.

$$P(8, 3) = \frac{8!}{5!} = 8(7)(6) = 336$$

Client Visits

Delta Services can visit three of seven clients before noon. Each client is visited at most once. How many different visit sequences are possible?

$$P(7, 3) = \frac{7!}{4!} = 7(6)(5) = 210$$

Presentation Schedule

All four projects, Alpha, Beta, Gamma, and Delta, will be presented once. How many presentation orders are possible?

Permutations: Examples

Here all four projects are arranged, so $r = n = 4$. Using $0! = 1$:

$$P(4, 4) = \frac{4!}{0!} = 4(3)(2)(1) = 24$$

Combinations: Examples

Project Selection

Beta Energy can fund three of seven projects. All require the same investment amount. Count the selected groups; selecting the same three projects in a different order does not create a new group.

$$C(7, 3) = \frac{7!}{3!4!} = \frac{7(6)(5)}{3(2)(1)} = 35$$

Audit Sample

Gamma Audit selects four different invoices from twelve for review. Only which invoices are selected matters; the order of review is ignored.

$$C(12, 4) = \frac{12!}{4!8!} = \frac{12(11)(10)(9)}{4(3)(2)(1)} = 495$$

Lottery

A lottery draws five different integers from 1 to 40. A result is the set of five numbers; their order is ignored.

$$C(40, 5) = \frac{40!}{5!35!} = \frac{40(39)(38)(37)(36)}{5(4)(3)(2)(1)} = 658,008$$

Combinations: Tablet Selection

A store has the following tablets:

Brand	Number of tablets
Apple	8
Samsung	6
Huawei	4
Total	18

Select three tablets without replacement. Count individual physical tablets as distinct; selection order does not matter.

- How many selections are possible?

$$C(18, 3) = \frac{18!}{3!15!} = \frac{18(17)(16)}{3(2)(1)} = 816$$

- How many selections contain exactly two Apple tablets and one Samsung tablet?

Choose two of the eight Apple tablets:

$$C(8, 2) = \frac{8!}{2!6!} = \frac{8(7)}{2(1)} = 28$$

Combinations: Tablet Selection

Choose one of the six Samsung tablets:

$$C(6, 1) = \frac{6!}{1!5!} = 6$$

Each of the 28 Apple pairs can be combined with any of the six Samsung tablets.

$$C(8, 2)C(6, 1) = 28(6) = 168$$

- How many selections contain tablets from exactly two brands?

For each brand pair, select two tablets from one brand and one from the other.

Brands represented	Number of selections
Apple and Samsung	$C(8, 2)C(6, 1) + C(8, 1)C(6, 2) = 288$
Apple and Huawei	$C(8, 2)C(4, 1) + C(8, 1)C(4, 2) = 160$
Samsung and Huawei	$C(6, 2)C(4, 1) + C(6, 1)C(4, 2) = 96$

$$\text{Total} = 288 + 160 + 96 = 544$$

Exercise 1

A firm has six candidates.

- How many three-person shortlists are possible?
- How many contain one specified candidate?
- Separately, how many lead-and-backup assignments are possible from all six?

Exercise 2

Gamma Audit will review three different invoices.

Sales channel	Available invoices
Online	O_1, O_2, O_3, O_4
Store	S_1, S_2, S_3

A complete review plan specifies the three invoices in review order, one of two auditors, and one of three available days. Either auditor can review any invoices on any day.

- How many different groups of three invoices can be selected?
- How many complete review plans are possible?
- How many complete plans include both sales channels and review an online invoice first?
- For a fixed plan, record whether each invoice contains an error (E) or no error (N), in review order. List the sample space, assuming all patterns are possible. How many outcomes remain if only the number of invoices containing errors is recorded? Does counting the patterns establish that they are equally likely?

Assigning Probabilities

Basis	Numerical example
Classical, equally likely outcomes	One face of a fair die: $1/6$
Relative frequency	45 complaints in 200 orders: $45/200 = 0.225$
Subjective assessment	An analyst assigns a 0.30 chance to approval

- Every assigned probability lies between 0 and 1.
- Probabilities across the full sample space sum to 1.

Probability of an Event

For sample points ω_i in $\mathcal{S}$ and an event $A \subseteq \mathcal{S}$:

$$P(\omega_i) \geq 0, \quad \sum_{\omega_i \in \mathcal{S}} P(\omega_i) = 1, \quad P(A) = \sum_{\omega_i \in A} P(\omega_i)$$

If the N sample points are equally likely, $P(A) = |A|/N$. From n observed trials, the relative-frequency estimate is n_A/n .

Orders: Counts

A retailer records 200 orders.

Delivery	Complaint	No complaint	Total
Late	30	20	50
On time	15	135	150
Total	45	155	200

- Let L denote a late order and C a complaint.
- For row i and column j , with cell count n_{ij} and total n :

$$p_{ij} = \frac{n_{ij}}{n}, \quad p_{i\cdot} = \sum_j p_{ij}, \quad p_{\cdot j} = \sum_i p_{ij}$$

Delivery	C	C^c	Total
L	0.150	0.100	0.250
L^c	0.075	0.675	0.750
Total	0.225	0.775	1.000

- The interior cells are joint probabilities.
- The row and column totals are marginal probabilities.

Events and Their Probabilities

From the order table:

$$P(L) = \frac{50}{200} = 0.25, \quad P(C) = \frac{45}{200} = 0.225$$

Both events occur for 30 orders:

$$P(L \cap C) = \frac{30}{200} = 0.15$$

Event operations select cells of the crosstabulation:

Operation	Meaning	Selected order counts
$A \cap B$	Both events	$L \cap C : 30$
$A \cup B$	At least one event	$L \cup C : 30 + 20 + 15 = 65$
A^c	Event A does not occur	$L^c : 15 + 135 = 150$

Exercise 3

A bank reviewed 240 loan applications.

Application channel	Approved	Declined	Total
Online	24	76	100
Branch	84	56	140
Total	108	132	240

An analyst uses $24/100$ to report that 24% of all applications were both online and approved.

- Is this correct?
- Calculate the share of all applications that were both online and approved and the overall approval share. What does $24/100$ measure?

Complement Rule

- An event and its complement cover the full sample space without overlap.

$$P(A^c) = 1 - P(A)$$

- For the order table:

$$P(L^c) = 1 - 0.25 = 0.75, \quad P(C^c) = 1 - 0.225 = 0.775$$

Addition Rule

- For any two events,

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- Orders that were late or generated a complaint include 30 orders counted in both groups.

$$\begin{aligned} P(L \cup C) &= P(L) + P(C) - P(L \cap C) \\ &= 0.25 + 0.225 - 0.15 = 0.325 \end{aligned}$$

- Check by counting unique orders: $(50 + 45 - 30)/200 = 65/200 = 0.325$.

Mutually Exclusive Events

- Events A and B are mutually exclusive when $A \cap B = \emptyset$. Then,

$$P(A \cup B) = P(A) + P(B)$$

- Late and on-time delivery cannot occur for the same order:

$$P(L \cap L^c) = 0, \quad P(L \cup L^c) = P(L) + P(L^c) = 1$$

- Late delivery and a complaint can occur together.
- Their overlap must be subtracted in the addition rule.

Exercise 4

A retailer reviewed 600 shipments.

Delivery	Damaged	Not damaged	Total
Delayed	24	96	120
On time	36	444	480
Total	60	540	600

- The retailer gives one 30-MAD voucher for each shipment that was delayed or damaged.
- A spreadsheet budgets $30(120 + 60)$ MAD.
- Audit the budget. How many shipments had exactly one problem, and what proportion receive no voucher?

Conditional Probability

For $P(B) > 0$, $P(A | B) = \frac{P(A \cap B)}{P(B)}$.

The condition changes the denominator to the group known to have occurred.

Among the 50 late orders, 30 generated complaints. Among the 150 on-time orders, 15 generated complaints.

$$P(C | L) = \frac{P(L \cap C)}{P(L)} = \frac{30}{50} = 0.60$$

$$P(C | L^c) = \frac{P(L^c \cap C)}{P(L^c)} = \frac{15}{150} = 0.10$$

The full-sample calculation $30/200 = 0.15$ is $P(L \cap C)$: the probability that an order was both late and associated with a complaint.

Reversing the Condition

For events A and B with positive probabilities:

$$P(A | B) = \frac{P(A \cap B)}{P(B)}, \quad P(B | A) = \frac{P(A \cap B)}{P(A)}$$

For the orders:

$$P(L | C) = \frac{0.15}{0.225} \approx 0.667, \quad P(C | L) = \frac{0.15}{0.25} = 0.60$$

$P(L | C)$ describes late deliveries among complaints; $P(C | L)$ describes complaints among late deliveries.

Multiplication Rule

From the definition of conditional probability, for $P(A) > 0$:

$$P(B \mid A) = \frac{P(A \cap B)}{P(A)}$$

Multiply both sides by $P(A)$:

$$P(A)P(B \mid A) = P(A \cap B)$$

For the order table:

$$P(L \cap C) = P(L)P(C \mid L) = 0.25(0.60) = 0.15$$

Multiplication Rule: Example

A batch has 10 invoices: 3 overdue and 7 current. Draw two uniformly at random, one after the other, without replacement.

- What is the probability that both are overdue?

Let A denote an overdue first selection and B an overdue second selection.

After an overdue invoice is selected first, 2 of the remaining 9 are overdue.

$$P(A) = \frac{3}{10}, \quad P(B | A) = \frac{2}{9}$$

$$P(A \cap B) = P(A)P(B | A) = \frac{3}{10} \frac{2}{9} = \frac{1}{15} \approx 0.0667$$

- Check by counting first-second draw sequences.
 - There are $10(9) = 90$ possible sequences, and $3(2) = 6$ have both invoices overdue.

$$P(\text{both overdue}) = \frac{6}{90} = \frac{1}{15}$$

Partition of the Sample Space

A partition separates all outcomes into nonoverlapping groups.

- For events $A_1, \dots, A_k$ with $P(A_i) > 0$:

$$A_i \cap A_j = \emptyset \quad (i \neq j), \quad A_1 \cup \dots \cup A_k = S$$

- Any event B can be split into disjoint parts: $B = \bigcup_{i=1}^k (B \cap A_i)$.

Let $S = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ and $B = \{2, 4, 6, 8, 10\}$.

Partition group	Elements	Elements also in B
A_1	$\{1, 2, 3, 4\}$	$\{2, 4\}$
A_2	$\{5, 6, 7, 8, 9, 10\}$	$\{6, 8, 10\}$

$A_1 \cap A_2 = \emptyset$, $A_1 \cup A_2 = S$, and the two parts of B contain $2 + 3 = 5$ elements.

Late (L) and on-time (L^c) orders form a partition.

- Every order is in exactly one group.
- Therefore every complaint is in exactly one of these two parts:

$$C = (C \cap L) \cup (C \cap L^c)$$

Total Probability

The two parts $C \cap L$ and $C \cap L^c$ cannot occur together, so their probabilities add:

$$P(C) = P(C \cap L) + P(C \cap L^c)$$

Apply the multiplication rule to each part:

$$\begin{aligned} P(C) &= P(L)P(C \mid L) + P(L^c)P(C \mid L^c) \\ &= 0.25(0.60) + 0.75(0.10) = 0.150 + 0.075 = 0.225 \end{aligned}$$

For any event B and partition $A_1, \dots, A_k$, the general result is

$$P(B) = \sum_{i=1}^k P(B \cap A_i) = \sum_{i=1}^k P(A_i)P(B \mid A_i)$$

Independence

Events A and B are independent when $P(A \cap B) = P(A)P(B)$. If $P(B) > 0$, this is equivalent to $P(A | B) = P(A)$.

If L and C were independent, knowing L would not change the probability of C .

$$P(C | L) = 0.60 \neq P(C) = 0.225$$

Equivalently,

$$P(L \cap C) = 0.15 \neq P(L)P(C) = 0.25(0.225) = 0.05625$$

Late delivery and complaints are not independent in these records.

In the ten-element example, assume each element is equally likely. For $A_1 = \{1, 2, 3, 4\}$ and $B = \{2, 4, 6, 8, 10\}$:

$$P(A_1 \cap B) = \frac{2}{10} = 0.20 = P(A_1)P(B) = \frac{4}{10} \frac{5}{10}$$

Thus A_1 and B are independent.

Independence and Mutual Exclusiveness

L and L^c are mutually exclusive:

$$P(L \cap L^c) = 0$$

They are not independent:

$$P(L)P(L^c) = 0.25(0.75) = 0.1875 \neq 0$$

Mutual exclusiveness concerns overlap. Independence concerns whether one event changes the probability of the other.

In the same ten-element example, A_1 and A_2 are mutually exclusive, but

$$P(A_1 \cap A_2) = 0 \neq P(A_1)P(A_2) = \frac{4}{10} \frac{6}{10} = 0.24$$

so they are not independent.

Exercise 5

A campaign report claims that exposure and purchase are independent.

Group	Purchased	Did not purchase	Total
Exposed	48	72	120
Not exposed	16	64	80
Total	64	136	200

Assess the claim numerically. What does the table establish about the effect of the campaign?

Bayes' Theorem

Suppose $A_1, \dots, A_k$ partition the sample space, with $P(A_i) > 0$, and event B has occurred, with $P(B) > 0$. We want the probability of group A_i after observing B .

Start with conditional probability:

$$P(A_i | B) = \frac{P(A_i \cap B)}{P(B)}$$

The multiplication rule and total probability give

$$P(A_i \cap B) = P(A_i)P(B | A_i), \quad P(B) = \sum_{j=1}^k P(A_j)P(B | A_j)$$

Substituting both expressions:

$$P(A_i | B) = \frac{P(A_i)P(B | A_i)}{\sum_{j=1}^k P(A_j)P(B | A_j)}$$

- $P(A_i)$ is the prior probability.
- $P(B | A_i)$ is the chance of observing B within that group.
- $P(A_i | B)$ is the updated or posterior probability.

Bayes' Theorem: Orders

Delivery	Prior	Complaint rate	Joint probability
Late	0.25	0.60	$0.25(0.60) = 0.150$
On time	0.75	0.10	$0.75(0.10) = 0.075$
Total	1.00		$P(C) = 0.225$

$$P(L | C) = \frac{0.150}{0.225} \approx 0.667$$

Inspection Example

A screening system flags defective units.

Quantity	Probability
Unit is defective	0.10
Flag given defective	0.80
Flag given acceptable	0.20

What is the probability that a flagged unit is defective?

Inspection Example: Positive Flag

Expected counts for 1,000 units under the given probabilities:

Actual status	Flagged	Not flagged	Total
Defective	80	20	100
Acceptable	180	720	900
Total	260	740	1,000

$$P(\text{defective} \mid \text{flag}) = \frac{80}{260} \approx 0.308$$

The same result follows from $0.10(0.80)/[0.10(0.80) + 0.90(0.20)]$.

Inspection Example: No Flag

A unit without a flag can still be defective.

$$P(\text{defective} \mid \text{no flag}) = \frac{20}{740} \approx 0.027$$

Equivalently,

$$\frac{0.10(0.20)}{0.10(0.20) + 0.90(0.80)} \approx 0.027$$

The Base Rate

Keep the same inspection accuracy, but suppose the defect rate is 25 percent.

$$P(\text{defective} \mid \text{flag}) = \frac{0.25(0.80)}{0.25(0.80) + 0.75(0.20)} = \frac{0.20}{0.35} \approx 0.571$$

- With a 10 percent defect rate, the result was 0.308.
- The test did not change; the prior defect rate did.

Bayes' Theorem with Three Sources

A retailer receives shipments from three warehouses.

Warehouse	Share of shipments	Late rate
W_1	0.50	0.04
W_2	0.30	0.06
W_3	0.20	0.10

A shipment is late. Which warehouse is most likely to have sent it?

Three Sources: Calculation

$$P(L) = 0.50(0.04) + 0.30(0.06) + 0.20(0.10) = 0.058$$

Warehouse	Joint probability	Probability given late
W_1	0.020	$0.020/0.058 \approx 0.345$
W_2	0.018	$0.018/0.058 \approx 0.310$
W_3	0.020	$0.020/0.058 \approx 0.345$

W_1 and W_3 are equally likely sources of a late shipment, despite their different late rates.

Exercise 6

Two suppliers deliver the same component.

Supplier	Share of deliveries	Defect rate
A	0.70	0.02
B	0.30	0.08

A delivered component is defective. Calculate the probability that it came from B.

Exercise 7

A firm processes 1,000 orders.

- Four percent are fraudulent.
- A system flags 90 percent of fraudulent orders and 5 percent of legitimate orders.
- Reviewing a flagged order costs 15 MAD and prevents a 250 MAD loss if that order is fraudulent.
- Each fraudulent order not reviewed causes a 250 MAD loss.
- Compare the expected total cost of reviewing every flag with reviewing none.
- Should the firm review the flags?

What is next?

- Discrete Probability Distributions I
 - Random Variables
 - Developing Discrete Probability Distributions
 - Expected Value and Variance
 - Bivariate Distributions and Financial Portfolios
 - Reading(s): SBE Ch. 5.1 to 5.4

Exercise 8

An operations manager proposes investigating complaints only for late orders.

Delivery	Complaint	No complaint	Total
Late	35	65	100
On time	15	385	400
Total	50	450	500

Evaluate the proposal using the table. What should the manager investigate, and what cannot be concluded from these counts?

Exercise 9

An electronics manufacturer screens incoming components before assembly.

- Supplier records indicate that 1 percent are defective.
- The rapid test flags 95 percent of defective components and incorrectly flags 4 percent of acceptable components.
- A flagged component can undergo a manual check before it is used or discarded.

The operations manager proposes discarding every flagged component, claiming that a positive result makes a defect highly likely.

Evaluate this claim for a delivery of 10,000 components and recommend how to handle a positive result.

Exercise 10

A bank's audit team is reviewing six branches.

- Exactly two have elevated compliance risk, but the team does not know which ones.
- It randomly selects two branches for review.
- Before the detailed audit, a monitoring system checks the selected pair. It issues an alert with probability 0.80 if at least one selected branch is high risk, and 0.10 if neither is.

An alert occurs.

- The audit director wants to classify the pair as high risk without further verification, claiming that the alert proves one of the selected branches is high risk.
- Evaluate the claim and advise the audit team.